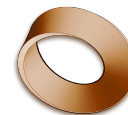


L2 – The Electromagnetic Spectrum: Overview

Unit 1: Light & EM Spectrum • MYP Sciences Year 8



Physics 8A and 8B • Mr. Rawson

8A	Mon 31 Aug, P2, 09:55–10:45	Next	Fri 4 Sep, P4 — 4 days
8B	Mon 31 Aug, P4, 12:15–13:05	Next	Fri 4 Sep, P2 — 4 days
Criterion	A — but see the red box	Lesson	2 of ≈ 12
To print	32 handouts (17 for 8A + 15 for 8B, one page each) and 10 card sheets. The cards are printed <i>once</i> : cut with 8A at P2, bagged, reused by 8B at P4. Scissors needed at P2 only.		
Groups	Pairs. 17 students makes 8 pairs and one three; 15 makes 7 pairs and one three. Ten card sets covers both with spares.		
Homework	None built, and the Class Log field is blank. [[ALEX: decide — whether to set any homework from this lesson]]		
Timing	Built to 40 min per the build brief; the Class Log clock spans read 50 (09:55–10:45, 12:15–13:05). [[ALEX: decide — which is real; if 50, the slack belongs in the sort and the debrief, not in more content]]		

No written Criterion A task — decide before Monday

Rebuilding the handout to the Clifton Hall single-spread layout **removed the three MYP command-term questions** (Explain, Describe, Analyse — Criterion A, strands A.i/A.iii), and the BBC Radio extension calculation went with them. As it stands the lesson produces **no written assessed evidence**: the filled spread is the only artefact. Note the asymmetry — 8B already banked a Criterion A write-up from the Beau Lotto lesson (28 Aug); 8A has *no* written assessed work yet this year.

[[ALEX: decide — no written Criterion A task since the Clifton Hall rebuild. Options: reinstate the three questions as a follow-on sheet, attach them to L3, or run L2 unassessed — but it is your call, and nothing has been invented here.]]

By the end of the lesson students can:

- name the seven regions of the electromagnetic spectrum, ordered by energy;
- place everyday uses (WiFi, sunburn, X-ray scanners, heat cameras...) in the right region;
- say that visible light is one narrow band of one continuous thing, and that the only difference between a radio wave and a gamma ray is the energy it carries.

Timing — run for 8A at P2; 8B deltas below

Time	Phase	What happens
0–3	Settle + launch	Pairs. Card sheets and scissors out. Do not name the lesson — no title on the board, and keep this plan out of sight. The sheet’s footer says only “Year 8 Physics • Unit 1” by design.
3–14	Cut and sort	Pairs cut the 19 pieces and sort them into groups that make sense to <i>them</i> — the card sheet’s own brief: think big, be creative, no single right answer yet. Circulate asking “why that grouping?” Steer nobody toward the answer.
14–19	Defend	Each pair names one of its groups and the reason. Collect category names on the board (medicine, danger, communication...) — they are needed for the debrief contrast. Nine groups in 8A, so hold each to ~30 s.
19–24	The reveal	Hand out the spread. 1 Across, (15, 8): “ <i>the manipulation of this hidden phenomenon is the single most important discovery in human history.</i> ” Let pairs crack it, then fill ELECTROMAGNETIC SPECTRUM together. This is the moment the lesson exists for.
24–34	Re-sort against the real thing	Cards onto the lined placement area beside the band each use belongs to; fill the seven numbered blanks. The frequency/wavelength scale carries no region names — see the question below on how the names reach the page.
34–40	Debrief + pack	Say the two take-aways out loud: visible light is one narrow band of one continuous spectrum, and radio-to-gamma differ only in energy. Contrast the board’s categories with the physicist’s grouping. 8A: bag the ten cut sets — 8B reuses them at P4.

The open delivery question. The scale on the handout shows frequency and wavelength but names no regions, and the seven blanks have to be filled from somewhere. [[ALEX: decide — how the seven names reach the page: project the Clifton Hall deck’s spectrum slide, list them on the board after the reveal, or collect them from the class and order them together]]

8B deltas (P4). Sets arrive pre-cut in bags from P2 — no scissors, roughly five minutes back; default it into the sort and the defence round, and adjust from how P2 actually went (the Class Log says exactly this). 8B has done the Lotto write-up, so they have already met “seeing is not simple” and the three command terms; 8A has not, so skip any callback to it at P2.

Questions to ask

1. What do WiFi and sunburn have in common? (*Both are electromagnetic waves — they differ only in energy.*)
2. Why can't you see the microwaves cooking your food? (*Your eyes detect only the visible band — one narrow slice of the whole.*)
3. Which is the odd one out: radio waves, sound waves, light? (*Sound — the other two are EM. This exposes the radio-equals-sound confusion.*)
4. Your categories were “medicine” and “communication” — why does the physicist's grouping cut straight across them? (*Physics groups by energy, not by what humans use it for.*)
5. What is the *only* difference between a radio wave and a gamma ray?

Common misconceptions

- **“Radio waves are sound waves.”** The broadcast travels as an EM wave; the sound is made by the speaker at the end. Question 3 exists for this.
- **“Radiation means nuclear and dangerous.”** The light from a torch is radiation. Expect the word to carry fear into the sort.
- **Seven separate things.** It is one continuous spectrum; the boundaries are human-drawn labels, which is why the scale on the sheet runs unbroken from 10^6 to 10^{19} Hz.
- **“Bigger waves are stronger.”** Backwards — gamma has the shortest wavelength and the most energy. The scale on the sheet shows both axes, so point at it rather than asserting it.

Differentiation

Support: a pair stuck on the open sort gets a narrower ask — “find all the pieces about seeing things” — not a hint at the spectrum. During placement the diagram does the teaching; send them back to the scale before you answer.

Extend: the two blank cards — invent a use and place it. Before the reveal, challenge quick pairs to order their groups from most to least dangerous, then see whether their order survives the reveal.

After the lesson

- **At the end of P2, bag the ten cut sets.** P4 depends on them coming back intact.
- **8A runs first.** Whatever the sort and the reveal actually took at P2, re-cut the timings before 12:15.
- Flip both Class Log entries to **Taught**; note how far each set got and whether the reveal landed.
- The filled spread is the only evidence of learning this lesson produces — the Criterion A decision (red box, page 1) is still open, and L3 on Friday is the next chance to place a written task.
- Keep the board photo or a note of the categories the pairs invented — the danger/health groupings seed the Criterion D discussion when microwaves and ionising radiation come round (unit map, L8–9).